

Embedded Language Flows as a Constrained Latent Diffusion Model

Setup

Latent Diffusion Models (LDM). A Latent Diffusion Model in the sense of Rombach et al. (2022) is a triple

$$\mathcal{M}_{\text{LDM}} = (E_{\text{LDM}}, \mathcal{F}, D_{\text{LDM}}),$$

together with a training objective $\mathcal{L}_{\text{LDM}}$, where:

1. $E_{\text{LDM}}: \mathcal{X} \rightarrow \mathcal{Z}$ is a (possibly learned) encoder mapping a data sample $x \in \mathcal{X}$ to a latent $z \in \mathcal{Z} \subseteq \mathbb{R}^d$.
2. $\mathcal{F}$ is a stochastic process on $\mathcal{Z}$ that progressively transports a target latent $z_1 = E_{\text{LDM}}(x)$ to a tractable prior latent $z_0 \sim p_0$. Concretely, $\mathcal{F}$ is specified by a family of marginals $\{p_t(z_t \mid z_1)\}_{t \in [0,1]}$ and a parametric vector field or score network $v_\theta(z_t, t)$ trained so that the induced ODE/SDE pushes p_0 onto the data latent distribution $E_{\text{LDM}\#} p_{\text{data}}$.
3. $D_{\text{LDM}}: \mathcal{Z} \rightarrow \mathcal{X}$ is a decoder that maps a sampled latent $\hat{z}$ back to the data space. In Rombach et al. this is a separate decoder network trained jointly with E_{LDM} via a (VQ-)VAE-style reconstruction objective.
4. The training objective decomposes as

$$\mathcal{L}_{\text{LDM}} = \mathcal{L}_{\text{latent}}(\theta; \mathcal{F}) + \lambda \mathcal{L}_{\text{recon}}(E_{\text{LDM}}, D_{\text{LDM}}),$$

where $\mathcal{L}_{\text{latent}}$ supervises the diffusion process in $\mathcal{Z}$ (denoising score matching, ε -prediction, flow matching, etc.) and $\mathcal{L}_{\text{recon}}$ supervises the encoder/decoder pair against samples in $\mathcal{X}$.

We emphasise that the LDM framework is agnostic to the specific noising process: the original work uses DDPM-style Gaussian diffusion, but continuous-time Flow Matching (Lipman et al. 2023) is a strictly valid choice of $\mathcal{F}$ since it specifies the same object—a parametric transport on $\mathcal{Z}$.

Embedded Language Flows (ELF). ELF (Hu et al. 2026) is a generative model over discrete token sequences. Let $\mathcal{V}$ be a finite vocabulary of size $|\mathcal{V}|$, and let $W_E \in \mathbb{R}^{d \times |\mathcal{V}|}$ be the input token embedding matrix of a Transformer language model. For a token $x \in \mathcal{V}$ with one-hot encoding $\mathbf{1}_x \in \mathbb{R}^{|\mathcal{V}|}$, write $W_E x := W_E \mathbf{1}_x \in \mathbb{R}^d$. ELF defines:

1. An *encoder* $E_{\text{ELF}}: \mathcal{V} \rightarrow \mathbb{R}^d$ given by the linear map $E_{\text{ELF}}(x) = W_E x$, with W_E *frozen* at the value taken from the pretrained language model.

2. A *flow-matching* diffusion in continuous embedding space $\mathcal{Z} = \mathbb{R}^d$, with linear interpolant $z_t = (1-t)z_0 + tz_1$, $z_0 \sim \mathcal{N}(0, I)$, $z_1 = E_{\text{ELF}}(x)$. The conditional vector field $u_t(z_t | z_1) = z_1 - z_0$ is regressed by a parametric $v_\theta(z_t, t)$ via the flow-matching loss

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, z_0, z_1} \|v_\theta(z_t, t) - (z_1 - z_0)\|_2^2.$$

3. A *decoder* $D_{\text{ELF}}: \mathbb{R}^d \rightarrow \Delta^{|\mathcal{V}|-1}$ given by the weight-tied softmax classifier

$$D_{\text{ELF}}(z) = \text{softmax}(W_E^\top z).$$

No new parameters are introduced for the decoder.

4. A *discretisation loss* at the final timestep $t = 1$ given by cross-entropy against the ground-truth token,

$$\mathcal{L}_{\text{CE}} = \mathbb{E}_x \left[-\log[D_{\text{ELF}}(z_1)]_x \right].$$

The total ELF objective is $\mathcal{L}_{\text{ELF}} = \mathcal{L}_{\text{FM}} + \lambda \mathcal{L}_{\text{CE}}$.

Proposition (ELF is a constrained LDM)

Proposition 1. *ELF is exactly the special case of LDM obtained by imposing the following four constraints on $\mathcal{M}_{\text{LDM}} = (E_{\text{LDM}}, \mathcal{F}, D_{\text{LDM}})$ and on $\mathcal{L}_{\text{LDM}}$:*

- (i) $E_{\text{LDM}}(x) = W_E x$ for a fixed (frozen) matrix $W_E \in \mathbb{R}^{d \times |\mathcal{V}|}$.
- (ii) $D_{\text{LDM}}(z) = \text{softmax}(W_E^\top z)$, with the same W_E as in (i) (weight tying).
- (iii) $\mathcal{F}$ is continuous-time Flow Matching with the linear Gaussian interpolant $z_t = (1-t)z_0 + tz_1$, $z_0 \sim \mathcal{N}(0, I)$.
- (iv) The reconstruction term $\mathcal{L}_{\text{recon}}$ is the cross-entropy of $D_{\text{LDM}}(z_1)$ against the input token x .

Conversely, every choice satisfying (i)–(iv) reproduces ELF exactly.

Proof

Proof. We verify each component of the LDM triple separately, then the loss, and finally combine.

Step 1 (Encoder). The LDM specification requires only that E_{LDM} be a measurable map $\mathcal{X} \rightarrow \mathcal{Z}$. Take $\mathcal{X} = \mathcal{V}$ (with the discrete sigma-algebra) and $\mathcal{Z} = \mathbb{R}^d$. The map $x \mapsto W_E \mathbf{1}_x$ is linear in the one-hot encoding and hence measurable, so it is a valid encoder. Constraint (i) is therefore an admissible specialisation: it pins E_{LDM} to a degenerate but allowed family. Freezing W_E amounts to removing E_{LDM} from the set of trainable parameters, which is a standard fine-tuning regime already covered by the LDM literature (e.g. frozen perceptual encoders in CLIP-conditioned LDMs).

Step 2 (Latent diffusion process). LDM requires a stochastic process $\mathcal{F}$ on $\mathcal{Z}$ together with a parametric vector field or score network v_θ trained so that pushforward of p_0 under the induced flow matches $E_{\text{LDM}} \# p_{\text{data}}$. Flow Matching with linear interpolant satisfies this requirement: its conditional vector field $u_t(z_t | z_1) = z_1 - z_0$ has a well-defined marginal $u_t(z_t) = \mathbb{E}[z_1 - z_0 | z_t]$, and the flow-matching loss $\mathcal{L}_{\text{FM}}$ is a Bregman regression that recovers u_t at its global minimiser (Lipman et al. 2023, Theorem 1). Integrating $\dot{z}_t = v_\theta(z_t, t)$ from $t = 0$ to $t = 1$ then transports

$p_0 = \mathcal{N}(0, I)$ to the latent data distribution $E_{\text{LDM}\#} p_{\text{data}}$. Hence constraint (iii) selects one valid instance of $\mathcal{F}$ from the LDM design space.

Step 3 (Decoder). LDM requires $D_{\text{LDM}}: \mathcal{Z} \rightarrow \mathcal{X}$. For $\mathcal{X} = \mathcal{V}$ the natural codomain of a probabilistic decoder is the simplex $\Delta^{|\mathcal{V}|-1}$ over $\mathcal{V}$, from which we sample. The map

$$z \mapsto \text{softmax}(W_E^\top z) \in \Delta^{|\mathcal{V}|-1}$$

is well defined, smooth, and surjective onto the relative interior of the simplex; followed by sampling, it gives a valid stochastic decoder $\mathcal{Z} \rightarrow \mathcal{V}$. Constraint (ii) further ties the decoder weights to those of the encoder. This tying is a hard parameter constraint of the form $W_D = W_E^\top$ on the otherwise free decoder weights $W_D \in \mathbb{R}^{|\mathcal{V}| \times d}$. Imposing such a constraint strictly reduces the LDM hypothesis class: every weight-tied decoder is an LDM decoder, but not vice versa. Crucially, the tying eliminates the $O(d \cdot |\mathcal{V}|)$ decoder parameters, since they coincide with those already counted in the encoder.

Step 4 (Loss). The LDM objective decomposes as $\mathcal{L}_{\text{LDM}} = \mathcal{L}_{\text{latent}} + \lambda \mathcal{L}_{\text{recon}}$. By Step 2, choosing $\mathcal{L}_{\text{latent}} = \mathcal{L}_{\text{FM}}$ is consistent with (iii). It remains to specify $\mathcal{L}_{\text{recon}}$. For a categorical output space, the canonical reconstruction loss is the negative log-likelihood, i.e. the cross-entropy between $D_{\text{LDM}}(z_1)$ and the input one-hot $\mathbf{1}_x$. This is precisely $\mathcal{L}_{\text{CE}}$, so constraint (iv) selects the canonical categorical reconstruction loss within the LDM template.

Step 5 (Combination). Substituting (i)–(iv) into the LDM definition yields

$$\mathcal{M}_{\text{LDM}}^* = \left(x \mapsto W_E x, \text{Flow Matching with linear interpolant}, z \mapsto \text{softmax}(W_E^\top z) \right),$$

with objective $\mathcal{L}_{\text{LDM}}^* = \mathcal{L}_{\text{FM}} + \lambda \mathcal{L}_{\text{CE}}$. Component-by-component, this matches Setup’s definition of ELF: encoder ($E_{\text{ELF}} = W_E$), latent process (flow matching with linear interpolant), decoder ($D_{\text{ELF}} = \text{softmax}(W_E^\top \cdot)$), and loss ($\mathcal{L}_{\text{ELF}} = \mathcal{L}_{\text{FM}} + \lambda \mathcal{L}_{\text{CE}}$). Hence $\mathcal{M}_{\text{LDM}}^*$ is ELF, and conversely every ELF model is obtained in this way. The equivalence is constructive: each constraint is fully specified, and the construction is invertible. $\square$

Consequences

1. **Toolkit transfer.** Because ELF is a constrained LDM, every technique developed for LDMs that does not rely on a separately learned decoder transfers immediately. Concretely:
 - *Classifier-free guidance* (Ho & Salimans 2022) is already used in ELF; the proposition shows this was not coincidental but inherited.
 - *Perceptual losses* of the form $\sum_\ell \|\phi_\ell(z_t) - \phi_\ell(z_t^*)\|^2$ for intermediate trajectory states, where ϕ_ℓ is a pretrained feature extractor, are admissible additions to $\mathcal{L}_{\text{ELF}}$ without altering the proposition.
 - *Joint encoder–decoder training with a KL regulariser* (the VAE branch of LDM) becomes available once constraint (i) is relaxed; see consequence 2.
2. **Relaxing the frozen encoder.** Constraint (i) is the strongest restriction. Replacing it with a trainable $W_E \in \mathbb{R}^{d \times |\mathcal{V}|}$ together with a KL penalty $\text{KL}(q_\phi(z | x) \parallel \mathcal{N}(0, I))$ recovers a categorical-input VAE-LDM, in which the embedding matrix is co-adapted to the diffusion latent geometry rather than inherited verbatim from a Transformer.

3. **Relaxing the weight-tied decoder.** Constraint (ii) imposes $W_D = W_E^\top$. The natural relaxation is a low-rank residual $W_D = W_E^\top + UV^\top$ with $U \in \mathbb{R}^{|\mathcal{V}| \times r}$, $V \in \mathbb{R}^{d \times r}$ and $r \ll \min(d, |\mathcal{V}|)$. This adds $O(r(d + |\mathcal{V}|))$ parameters while preserving the weight-tying inductive bias; this is the natural “ELF v2” candidate suggested by the embedding into the LDM family.

Discussion

What this proof gives. A constructive, invertible identification of ELF with a four-constraint specialisation of LDM. The identification is not metaphorical: every component of the LDM definition is matched component-wise, and the constraint set (i)–(iv) is exhaustive. As a structural result this licenses one-line citations of LDM theorems (e.g. guidance, latent regularisation) inside ELF analyses, and it pins down precisely which LDM hyperparameters ELF has fixed.

What this proof does not give. The equivalence is at the level of model class and objective, not at the level of empirical performance. It does *not* guarantee that any particular LDM technique will improve ELF in practice; that is an empirical question about how well the language-modelling inductive bias plays with the technique in question. Likewise, the proof says nothing about sample quality, training stability, or scaling behaviour—those require experiments.

How to test. Instrument an existing ELF training run with one LDM technique drawn from consequence 1 and measure a downstream metric. Concretely, the cleanest test is a perceptual loss on intermediate trajectory states: define ϕ_ℓ as the residual-stream activations of a frozen pretrained language model at layer ℓ , augment $\mathcal{L}_{\text{ELF}}$ with $\beta \sum_t \mathbb{E}_t \|\phi_\ell(z_t) - \phi_\ell(z_t^*)\|^2$ where z_t^* is the teacher trajectory, and compare downstream perplexity at matched compute. A positive result would corroborate the operational utility of the embedding into LDM; a null result would demonstrate that constraint (i) (the frozen, language-model-derived encoder) makes the latent space sufficiently structured that perceptual regularisation is redundant—itsself a useful negative finding.